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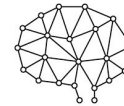
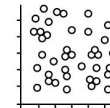
Sensor Data Evaluation From Industrial Wireless Sensor Network

CoDaS Project Course

Ahmad Ashraf Zargar, Raj Maharjan, Shreya Jindal, Malik Ahmad, July 16, 2025

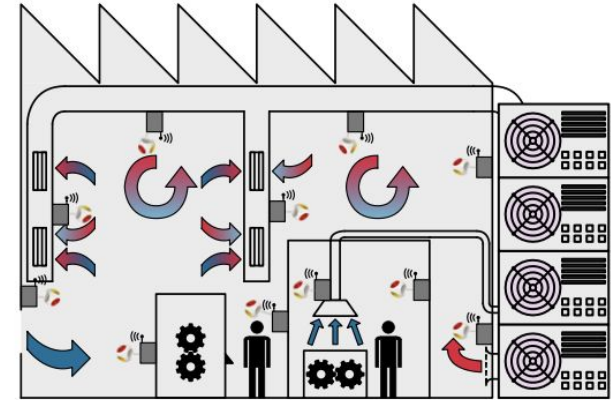
Agenda

- Introduction and Overview
- Exploratory Data Analysis
- Data Cleaning
- Disruption Analysis
- Task Classification: Unsupervised Learning
- Sensor/Node Analysis



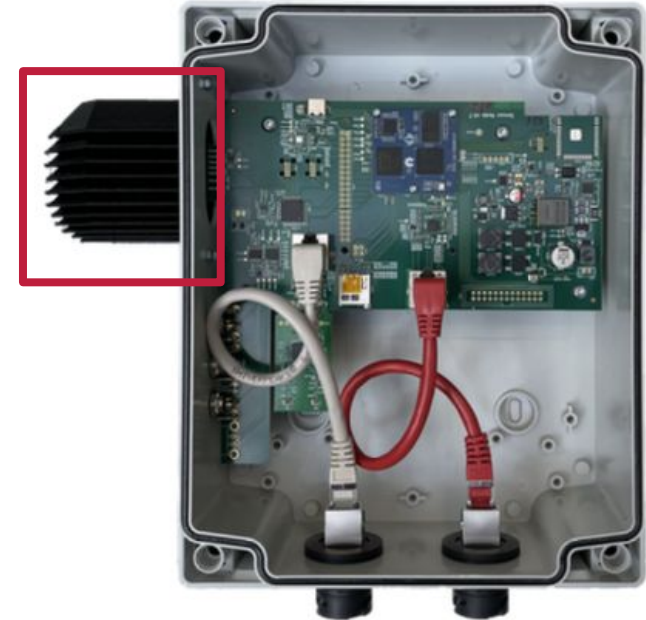
Industrial Wireless Sensor Network (IWSN)

- Systems of interconnected devices communicating wirelessly to monitor and control various industrial processes
- Monitor environmental conditions inside factories



Sensor Nodes

- 7 sensor nodes
- Microcontroller & High computing module
- External power supply
- Each sensor node has 3 sensors
 - Sensirion SCD30
 - Sensirion SPS30
 - Bosch BME680

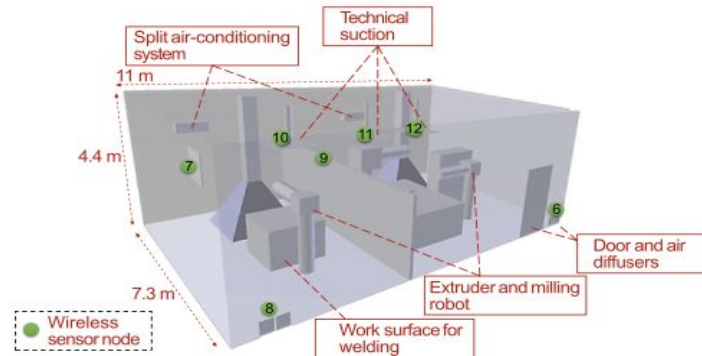
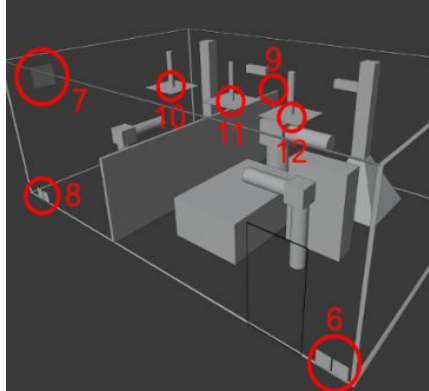


Sensors

- SCD30
 - Humidity
 - Temperature
 - Gas resistance
 - Pressure
 - CO2
- Bosch BME680
 - Humidity
 - Temperature
 - Gas resistance
 - Pressure
- SPS30
 - Mass Concentration of PM1 particles
 - Mass Concentration of PM2.5 particles
 - Mass Concentration of PM4 particles
 - Mass Concentration of PM10 particles
 - Number Concentration of PM0.5 particles
 - Number Concentration of PM01 particles
 - Number Concentration of PM02.5 particles
 - Number Concentration of PM4 particles
 - Number Concentration of PM10 particles

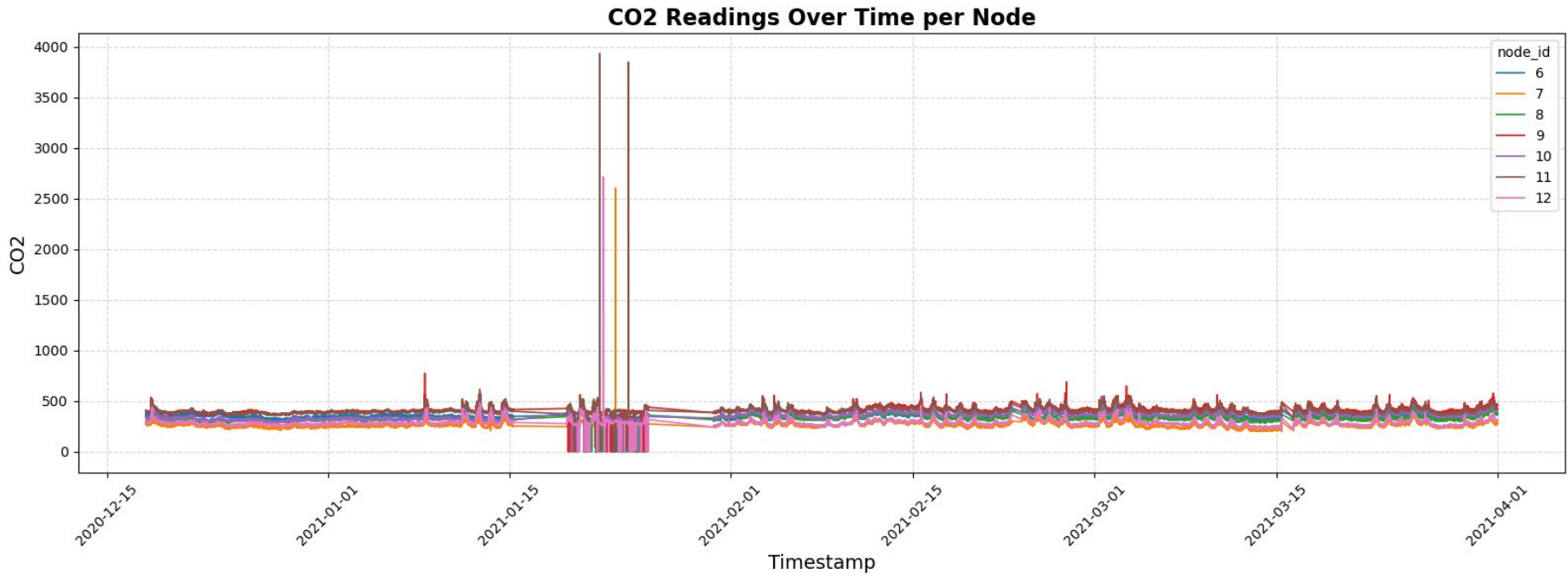
Integrated Manufacturing Lab

- 80qm size
- 3 technical suction
- 3 robot arms
- 3 doors
- Several small openings

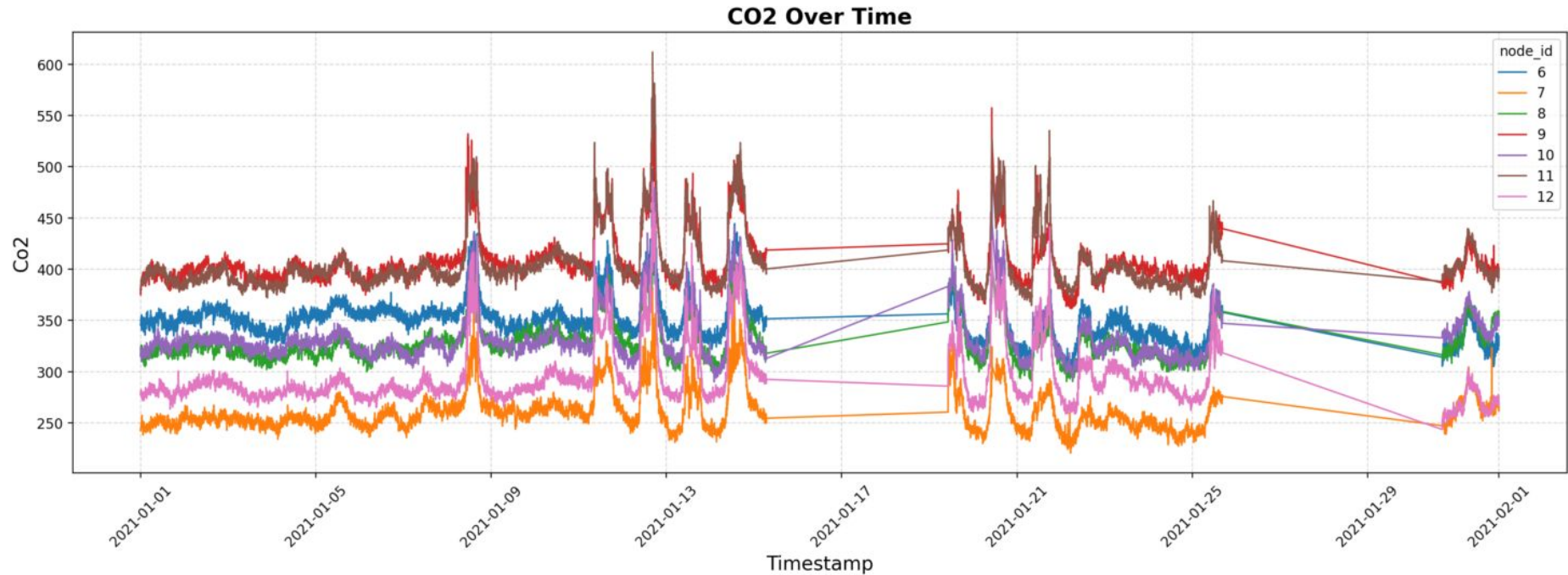


Exploratory Data Analysis

Exploratory Data Analysis



Exploratory Data Analysis



Exploratory Data Analysis

Inter | Intra Sensor Correlation

- Hourly CO2 reading Sensor SCD30.
- Hourly Gas Resistance reading BME680.
- Negatively Correlated relation between the sensor.

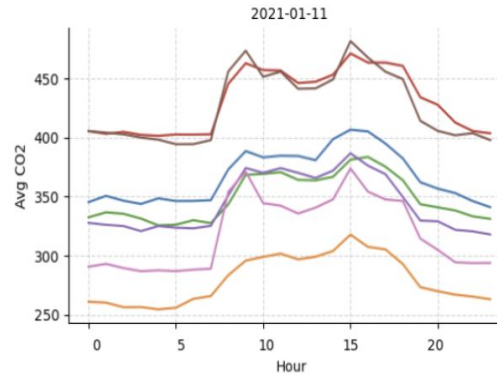


Fig Sensor-SCD30 CO2 (ppm)

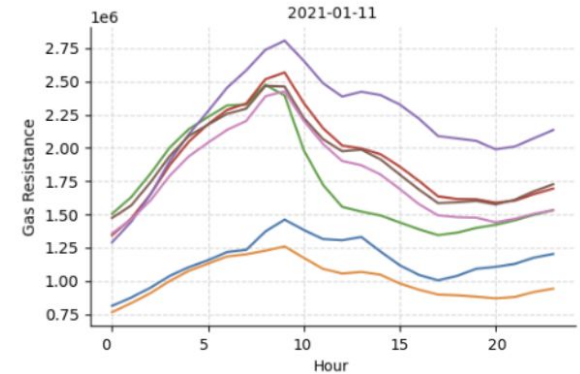
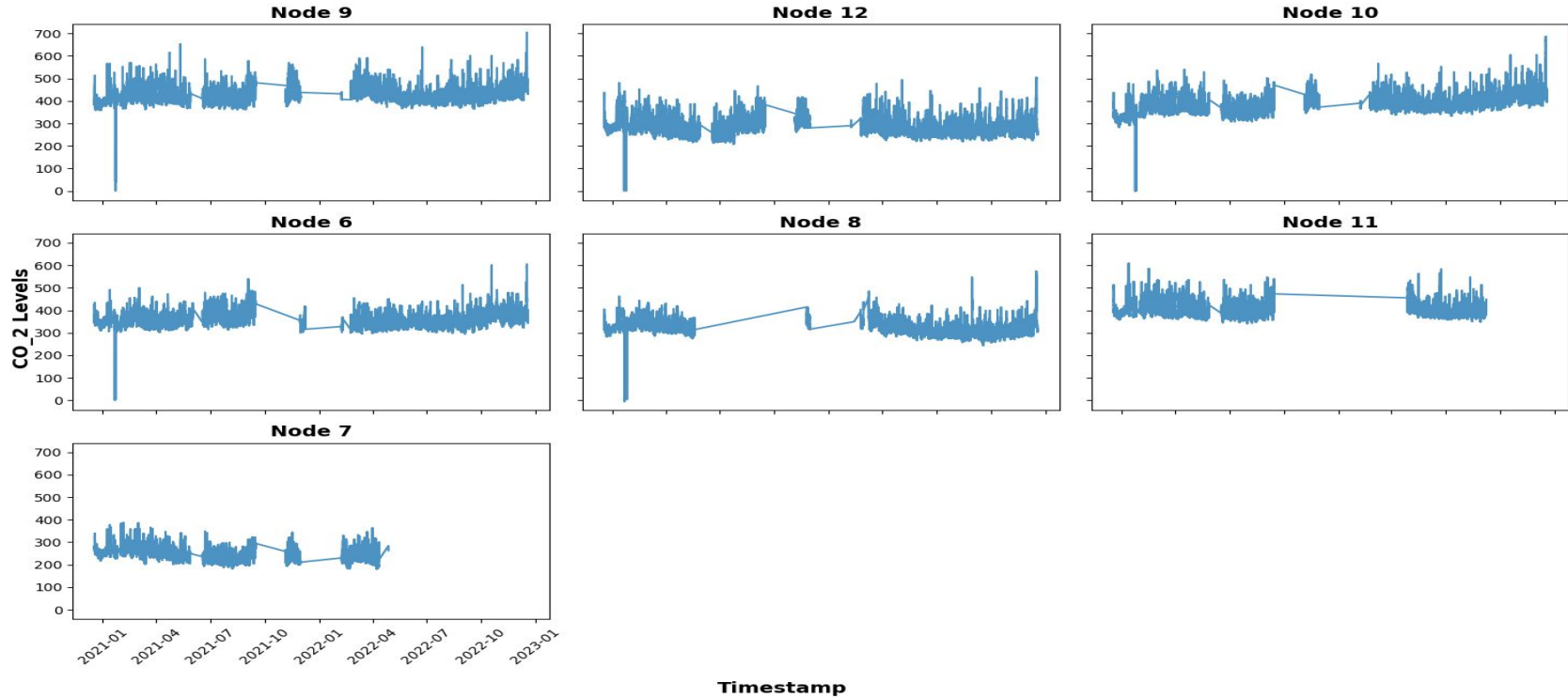


Fig Sensor-BME680 Gas Resistance (ohm)

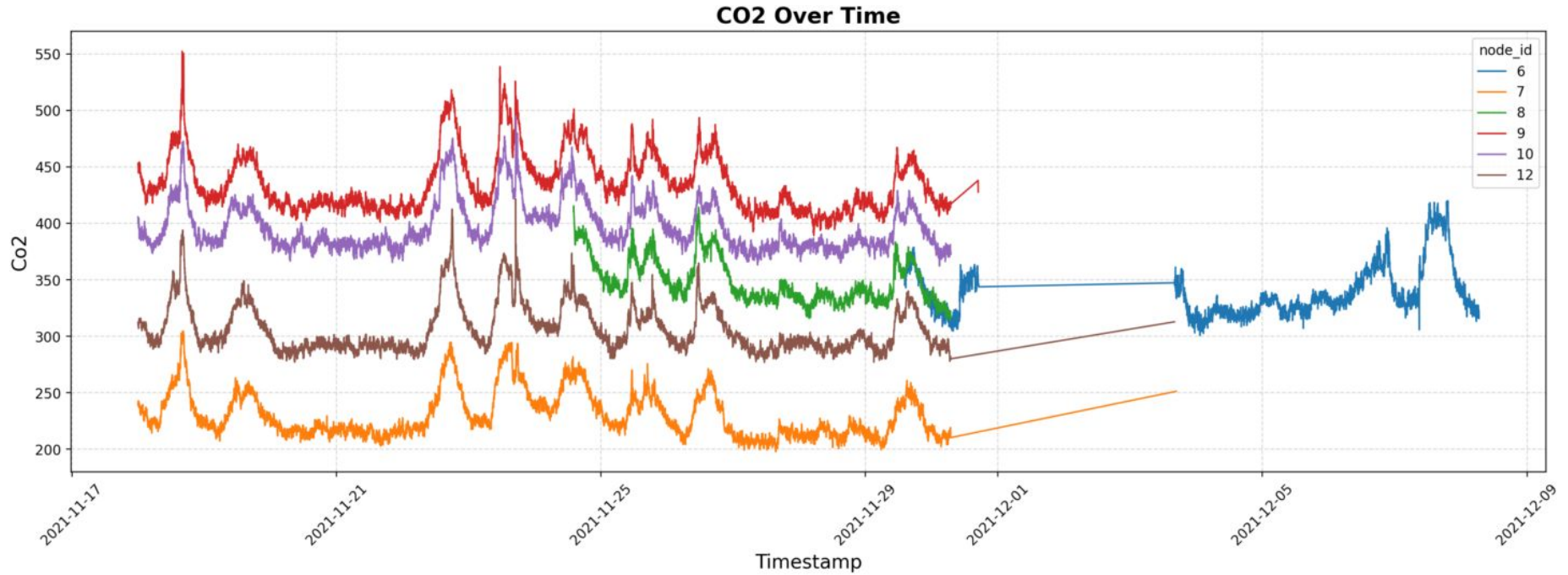
Fig Sensor-BME680 Gas Resistance (ohm)

Missing Time Periods (Node Level Disruption):

CO₂ Time Series for Different Nodes



Missing Time Periods (Node Level Disruption):

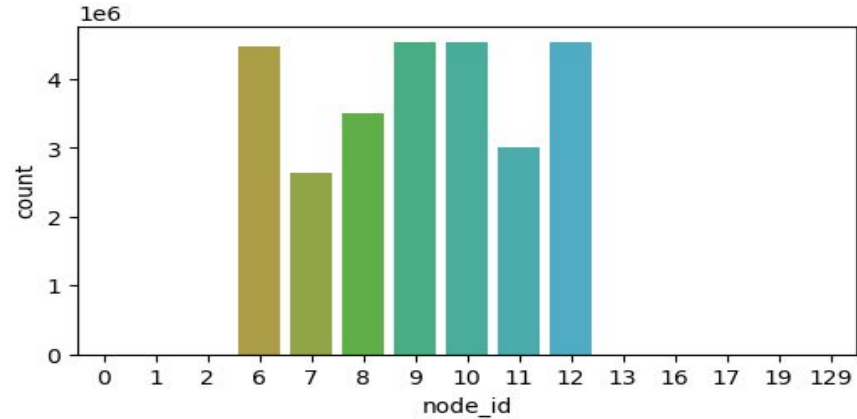


Data Cleaning and Sanitization

Data Cleaning and Sanitization

Node id of interest : 6 to 12

node_id	count(*)
0	306
1	1,636
2	306
6	4,484,374
7	2,647,905
8	3,510,181
9	4,548,516
10	4,539,850
11	3,017,305
12	4,545,636
13	24
16	7
17	6
19	1
129	1



Node Selection

Select node(s) to include

9 x 12 x 10 x 6 x 8 x 11 x 7 x

Available Node IDs in Dataset:

[10, 11, 9, 7, 12, 6, 8, 2, 1, 0, 13, 129, 19]

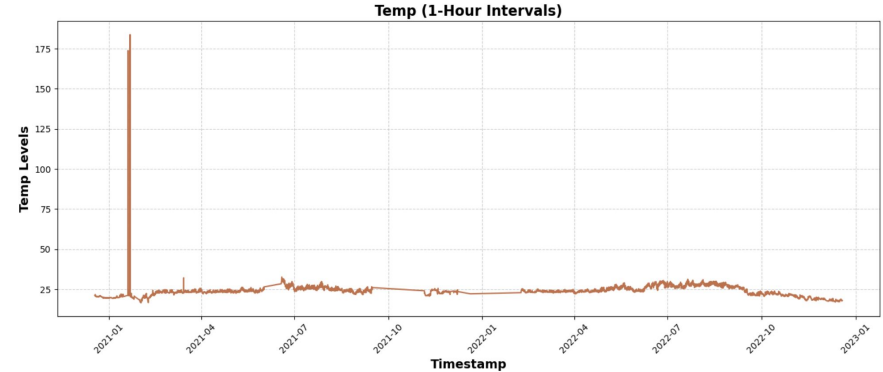
The following node(s) were excluded based on your selection:

[2, 1, 0, 13, 129, 19]

Data Cleaning and Sanitization

Outlier Analysis (Sensor-SCD30):

- Non-continuous rise and fall of readings.
- Keeping in mind the discrete notion of time, and non-uniformity over it.
- Sudden jumps or drops in sensor readings.



	id	node_id	timestamp	local_id	co2	temperature	humidity
1636010	17217157	10	2021-01-19 13:56:57	17216823.0	381.04	21.19	33.61
1636011	17217158	7	2021-01-19 13:56:58	17216824.0	277.03	22.03	31.64
1636012	17217159	11	2021-01-19 13:57:15	17216825.0	427.66	215622.34	0.00
1636013	17217160	12	2021-01-19 13:59:04	17216826.0	325.46	21.76	32.87
1636014	17217161	8	2021-01-19 13:59:07	17216827.0	377.81	20.94	34.80
1636015	17217162	11	2021-01-19 13:59:07	17216828.0	427.66	21.56	33.02

Data Cleaning and Sanitization

Hardcoded Bounds (Sensor-SCD30):

Feature	Upper Threshold	Lower Threshold
CO2 (ppm)	939.69	171.28
Temperature(°c)	33.61	16.06
Humidity(%r.H)	65.39	14.72

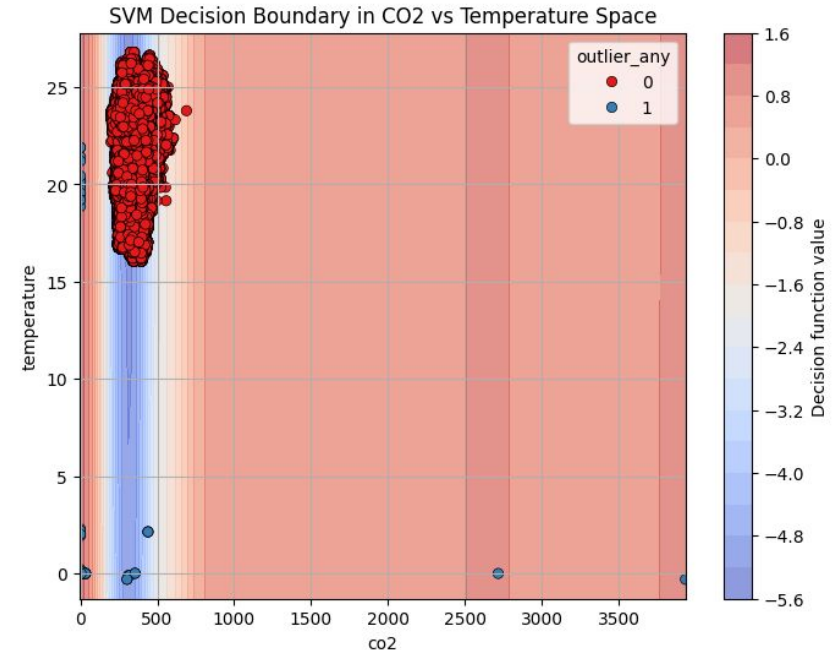
Hardcoded Bounds (Sensor-BME680):

Feature	Upper Threshold	Lower Threshold
Humidity(%r.H)	61.45	10.45
Temperature(°C)	33.61	13
Gas Resistance(ohm)	5430598	2988
Pressure (Pa)	103777	61521

Data Cleaning and Sanitization

SVM-Based (SCD30-BME680):

- Using the bounds from previous method, and training a binary classifier.
- Visual representation for decision boundaries.
- Computationally expensive, and require semantic understanding of data.
- Requires extensive data visualization.



Data Cleaning and Sanitization

Weighted IQR:

- User friendly approach.
- Dynamically filtering, with user input.
- Database independent.

Statistics Summary for Sensor-SCD30 Outlier readings (214 samples) [01.2021-02.2021]

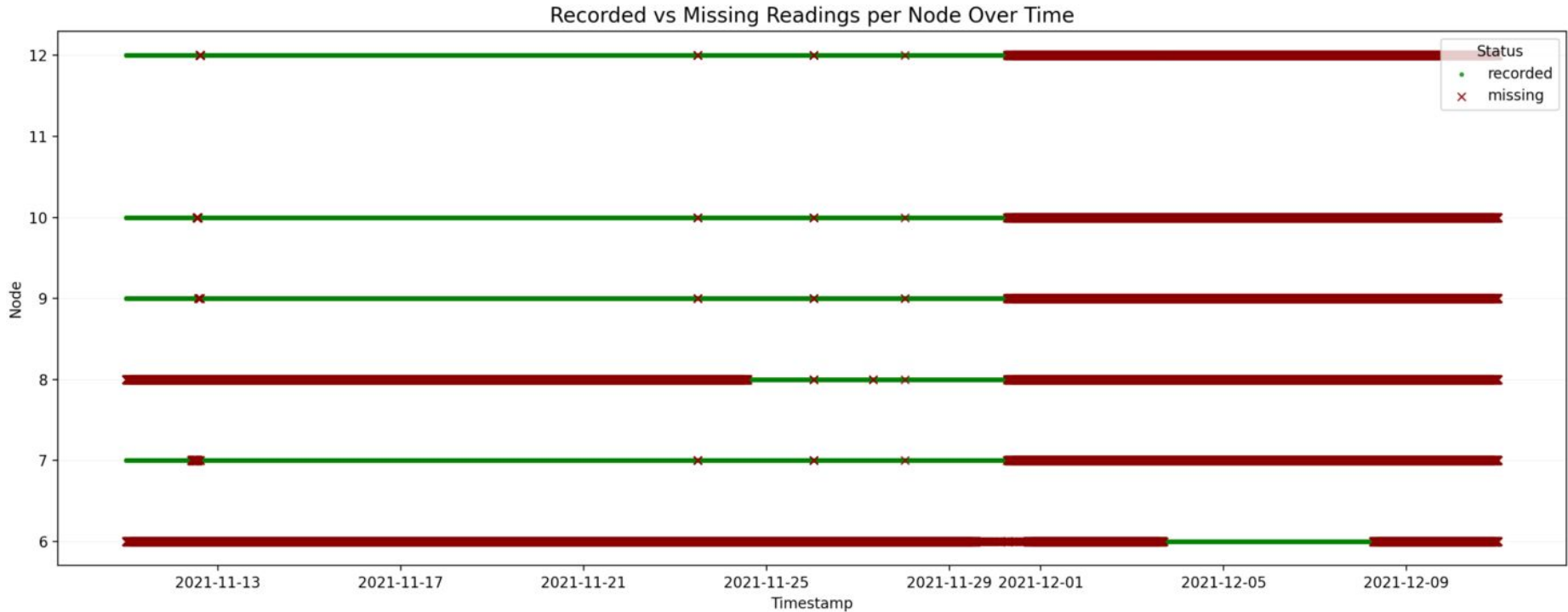
Statistic	CO ₂ (ppm)	Temperature (°C)	Humidity (%r.H.)
Count	214	214	214
Mean	166.15	2029.64	68.55
Min	-4.80	-4.50	-4.10
25%	-0.01	-0.01	0
50%	1.95	0	0
75%	38.45	0.01	0
Max	3932.90	215622.34	8700.22
Std Dev	558.97	20401.72	657

Disruption Analysis

Disruption Analysis

- Periods when sensor nodes stop recording data.
- Attributed by either: node failure, network failure, infrastructure failure, or holiday session.
- Disruptions also consists of different granularities.

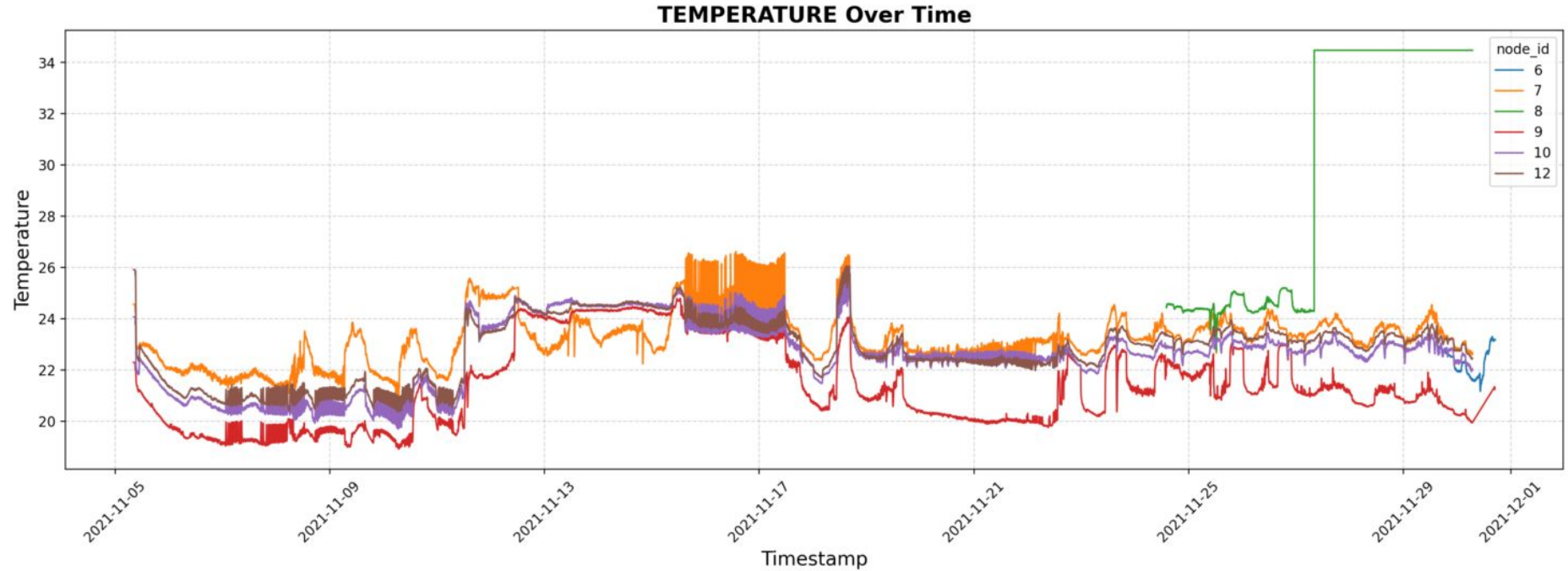
Disruption Analysis



Disruption Analysis

- Using expected interval of recordings; new metrics.
- Comparison of Uptime Vs. Downtime.
 - expected recording Vs. actual recording

Irregularity Analysis



Task Classification - TC 1-11

TC-1: Automated Pattern Recognition and Process State Detection

- **Challenge:** Understanding complex multi-sensor data patterns in real-time environments
- **Solution:** Unsupervised machine learning system that automatically discovers operational states from sensor data
- **Approach:** No manual labeling required - patterns emerge directly from sensor measurements

TC-3: Multi-Sensor Data Integration

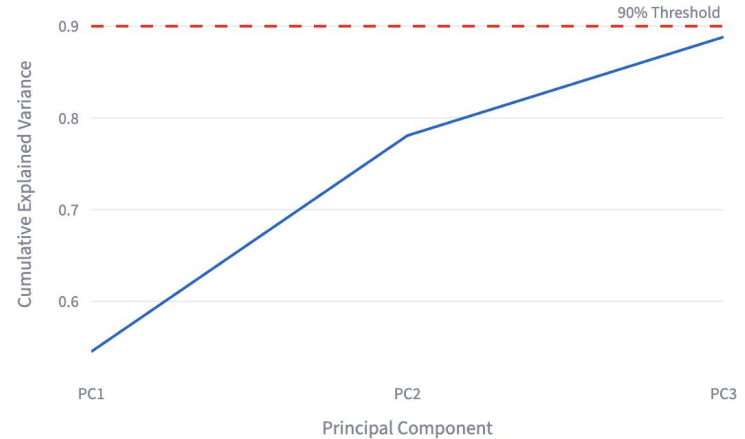
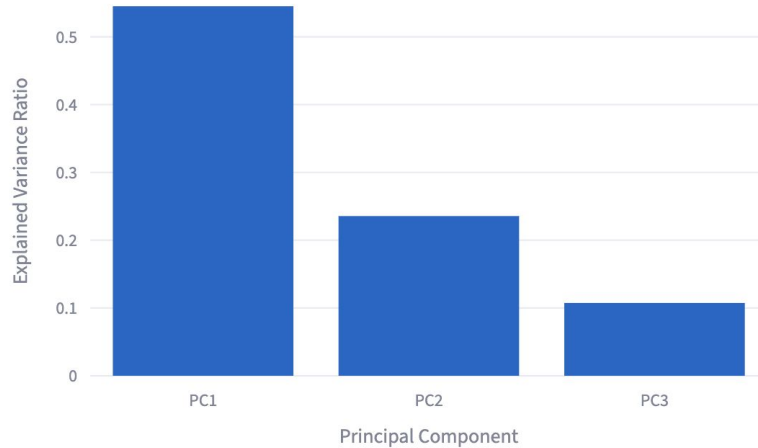
- **BME680**: Gas resistance, pressure, temperature, humidity (4 features)
- **SCD30**: CO2 concentration (1 feature)
- **SPS30**: Particulate matter mass/number concentrations, particle size (10 features)
- Total: 15 sensor features in standardized analysis

TC-4: Data Preprocessing Pipeline

- **Missing Value Handling:** Automatic detection and removal of incomplete measurements
- **Outlier Detection:** Statistical z-score filtering ($|z| \geq 3$) for anomaly removal
- **Feature Standardization:** StandardScaler normalization ensuring equal feature influence
- **Data Validation:** Comprehensive checks for data integrity and sensor functionality

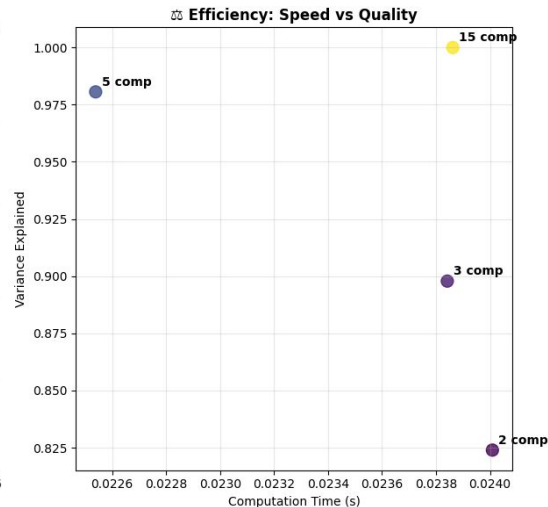
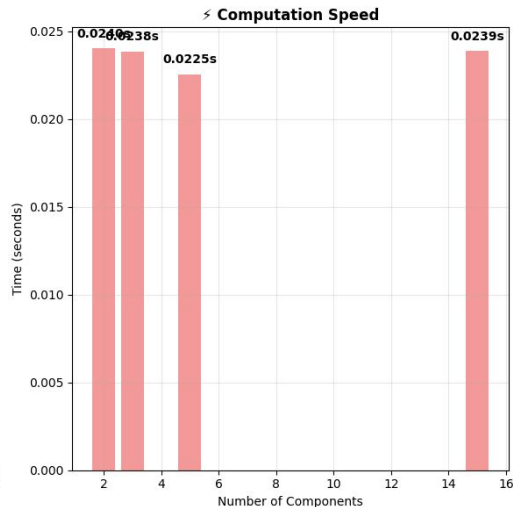
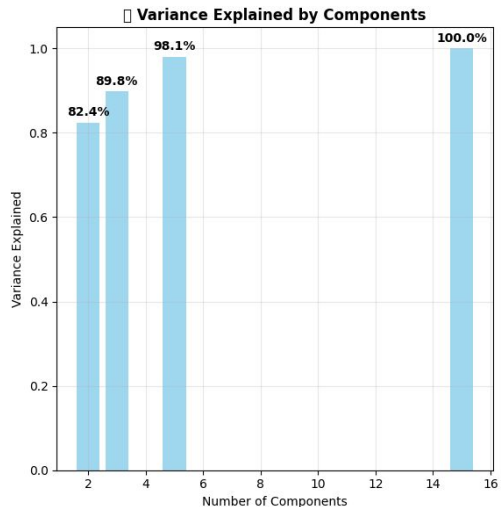
TC-5: Dimensionality Reduction & Visualization

- **15D** sensor space $\rightarrow$ 3D principal component space
- Explained variance analysis (typically 80-90% in 3 components)



TC-5B: Dimensionality Reduction & Visualization

- **15D** sensor space $\rightarrow$ 3D principal component space
- Explained variance analysis (typically 80-90% in 3 components)

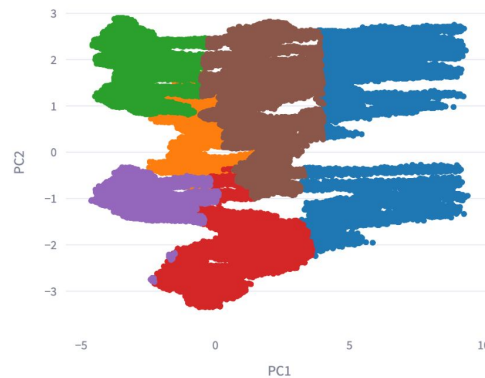


TC-6: Advanced Clustering Analysis

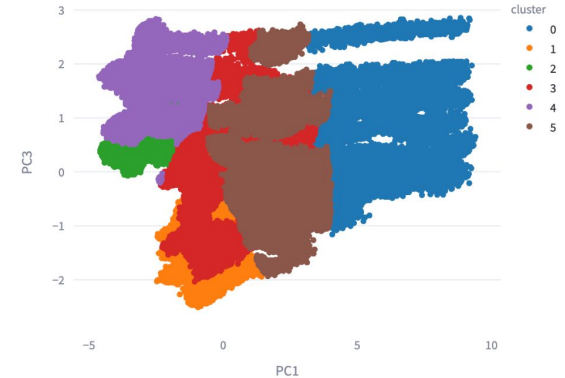
- Multiple perspective views (PC1 vs PC2, PC1 vs PC3)
- K-Means clusters
- Clustering Philosophy: "Let the Data Speak"
- **No Temporal Features:** No hour, day-of-week, or date information used in clustering

 Principal Component Analysis

Sensor States (PC1 vs PC2)



Sensor States (PC1 vs PC3)

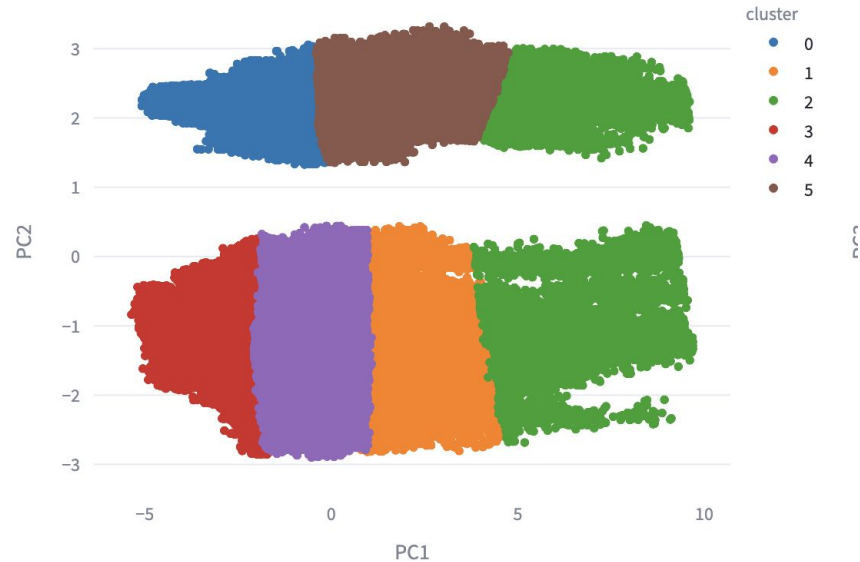


2022-11-16 to 2022-11-22

TC-6A: Advanced Clustering Analysis

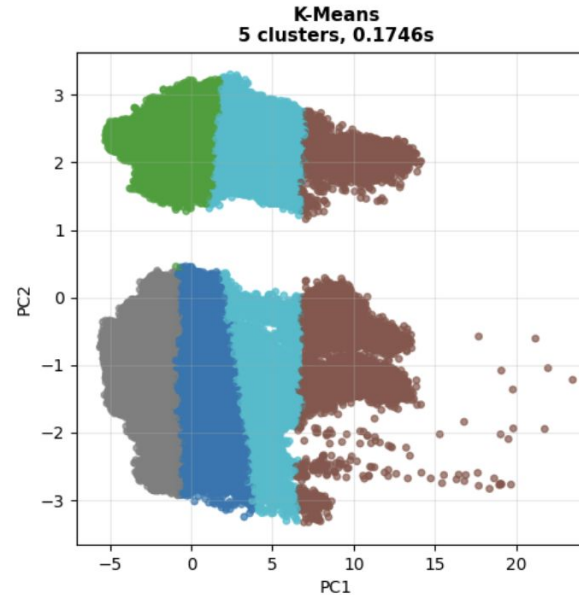
- K-Means clusters
- Date range: 2022-07-01 to 2022-07-07

Sensor States (PC1 vs PC2)



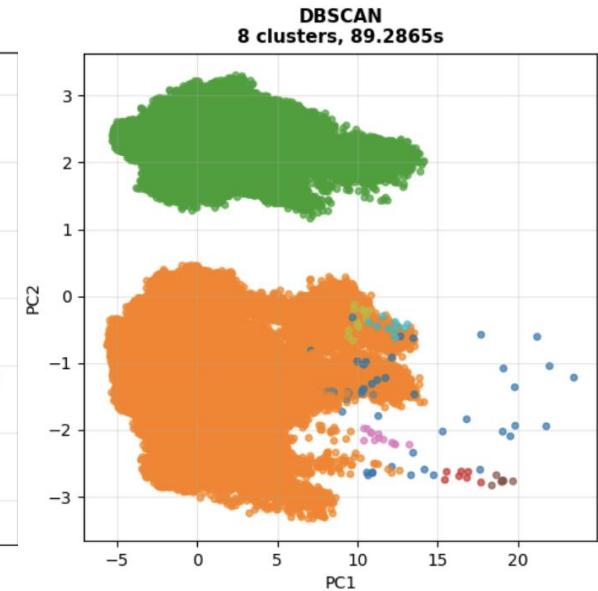
TC-6B: Advanced Clustering Analysis

- K-Means vs DBScan
- Date range:
2022-07-01 to
2022-07-07



K-Means Results:

- Clusters: 5
- Time: 0.1746s



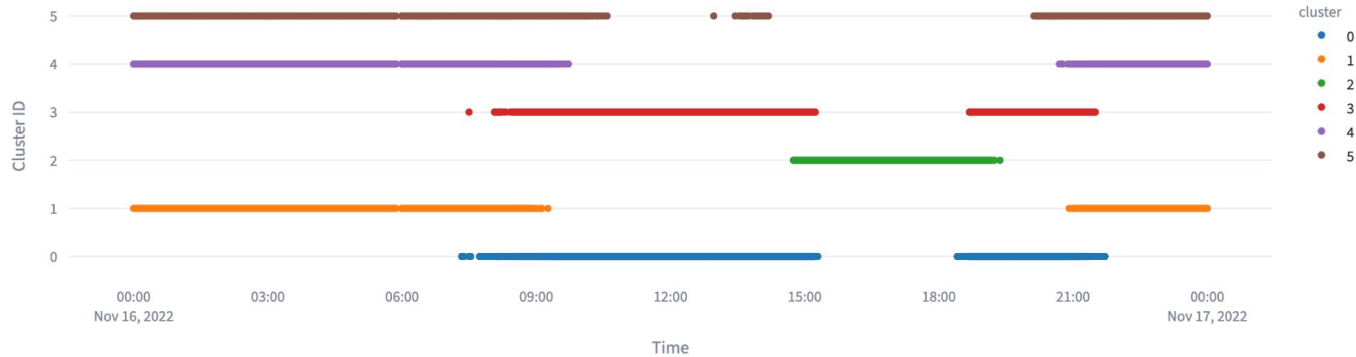
DBSCAN Results:

- Clusters: 8
- Time: 89.2865s
- Noise points: 38

TC-7: Temporal Analysis

- Clusters follow the hourly boundaries for a single day.
- No Temporal Features: No hour, day-of-week, or date information used in clustering

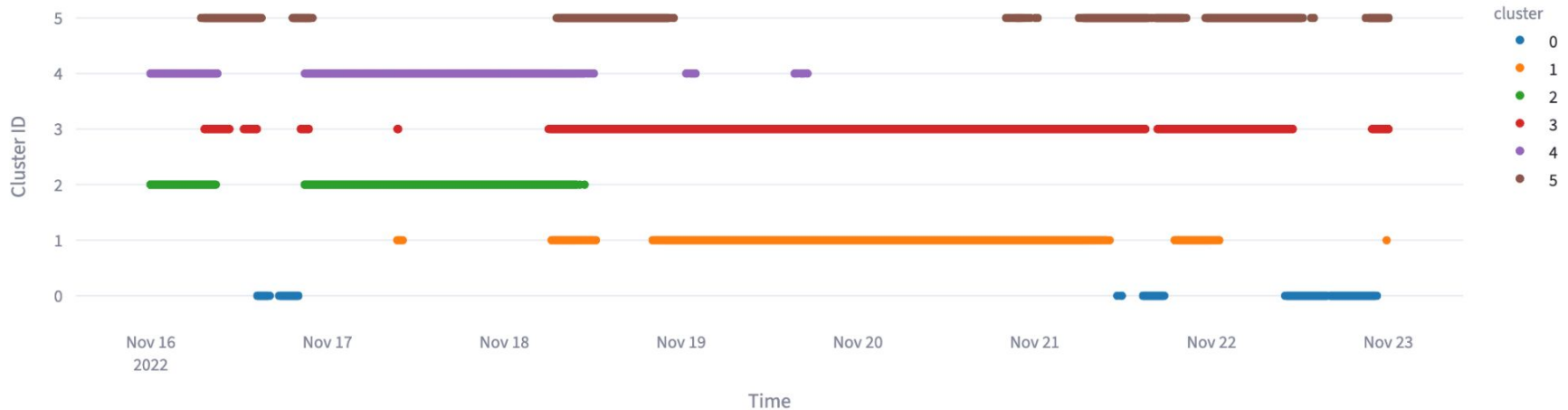
Process States Over 1 Days (2022-11-16 to 2022-11-16)



TC-8 B: Temporal Analysis

- Clusters for the whole week.

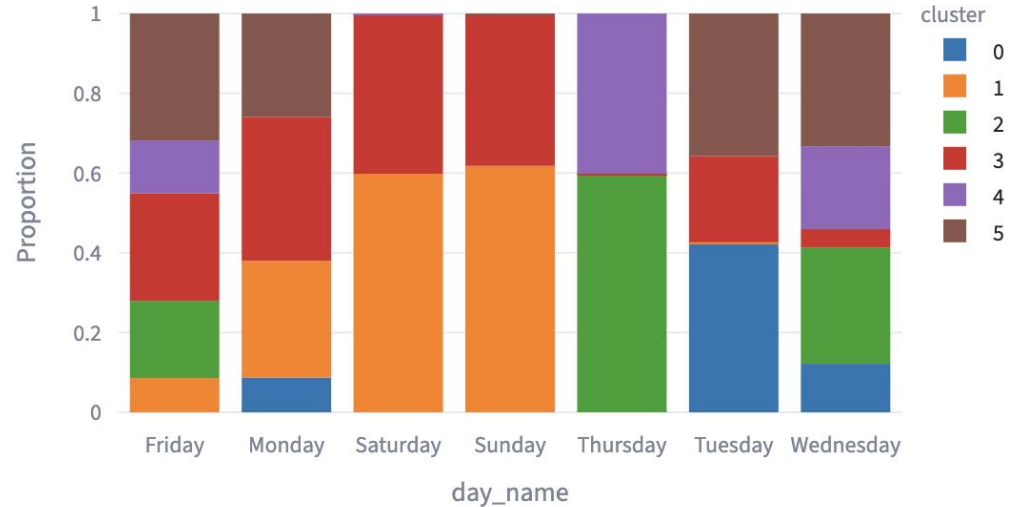
Process States Over 7 Days (2022-11-16 to 2022-11-22)



TC-8 C: Temporal Analysis

- Clusters by day of week.
- Weekends stand out.

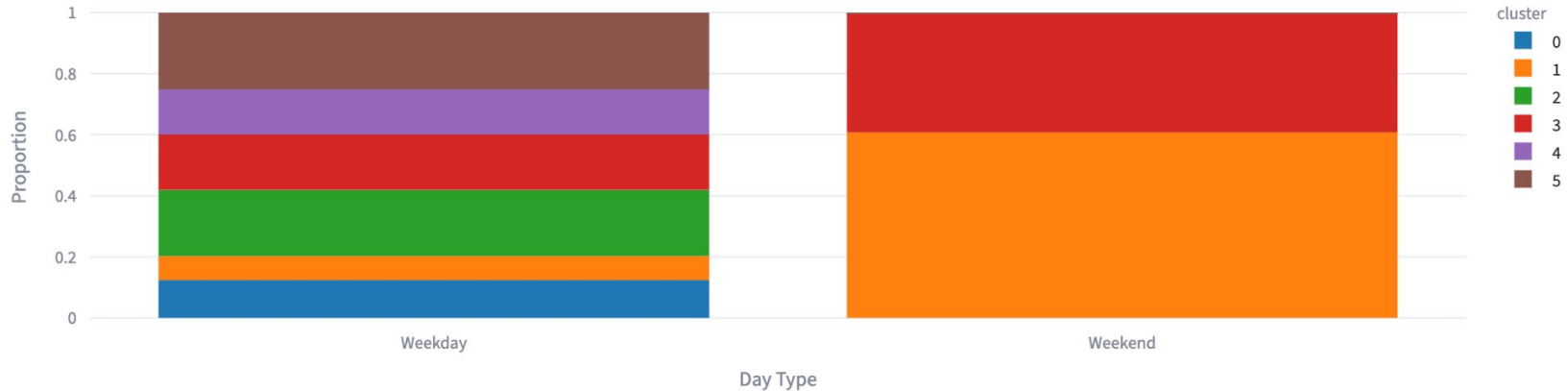
Normalized Cluster Distribution by Day of Week



TC-8 D: Temporal Analysis

- Clusters by day of week.
- Weekends stand out.

Cluster Distribution: Weekday vs Weekend



TC-9: Process Day Detection Methodology

- Intelligent Label Assignment
- **Primary Indicator:** CO2 levels during working hours (co2_working_mean)
- **Decision Logic:** Higher working-hour CO2 = Process Day classification
- Automatic Mapping: Clusters automatically labeled as "Process" or "Baseline"

 Process Day Detection Results

Total Days

7

Process Days

3 (42.9%)



Process Days

2022-11-16 (Wednesday), 2022-11-18 (Friday), 2022-11-22 (Tuesday)



Baseline Days

2022-11-17 (Thursday), 2022-11-19 (Saturday), 2022-11-20 (Sunday), 2022-11-21 (Monday)

TC-10: Cluster Distinctiveness Analysis

- Z-Score Deviation Mapping:
 $(\text{cluster_mean} - \text{overall_mean}) / \text{overall_std}$
- **Significance Threshold:**
 $|\text{z-score}| > 1.0$ for notable deviations
- **Direction Classification:** HIGH ($\uparrow$) vs. LOW ($\downarrow$) relative to baseline
- Top 5 Distinctive Features: Most characteristic sensors per operational state

Cluster 2

🎯 Cluster Distinctiveness

Notable deviations from overall average:

- **bme_hum:** $\uparrow$ HIGH ($+1.23\sigma$)
- **mass_conc_pm1:** $\downarrow$ LOW (-1.04σ)
- **mass_conc_pm2_5:** $\downarrow$ LOW (-1.03σ)
- **mass_conc_pm4:** $\downarrow$ LOW (-1.03σ)
- **mass_conc_pm10:** $\downarrow$ LOW (-1.03σ)

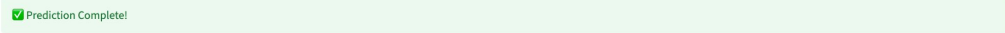


TC-11: Continuous Discovery Capabilities

- Incremental Pattern Learning:
New patterns discovered as more data becomes available
- Pattern Stability Monitoring:
Tracking whether discovered patterns remain consistent
- Seasonal Adaptation: Pattern evolution tracking across longer time periods
- Real-time Classification:
Immediate categorization of new data points into discovered patterns

Gas Resistance	506383	-	+
Bme Hum	2871	-	+
Mass Conc Pm2.5	1631	-	+
Num Conc Pm0.5	7214	-	+
Num Conc Pm4	12154	-	+
Pressure	9989332	-	+
Co2	29617	-	+
Mass Conc Pm4	1249	-	+
Num Conc Pm1	7074	-	+
Num Conc Pm10	12157	-	+
Bme Temp	1852	-	+
Mass Conc Pm1	1522	-	+
Mass Conc Pm10	1652	-	+
Num Conc Pm2.5	12146	-	+
Particle Size	53	-	+





Predicted Cluster	Confidence	Distance to Center
3	85.3%	1.02

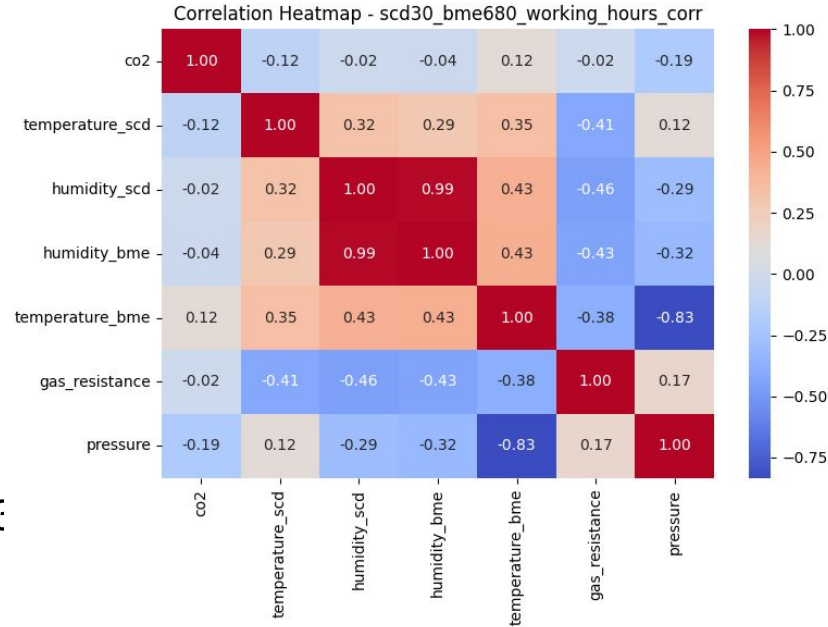
 Distance to All Clusters

Cluster	Distance	Predicted
3	1.015034	 YES
4	3.159769	No
1	3.404362	No
5	4.204297	No

Cross-Sensor Analysis

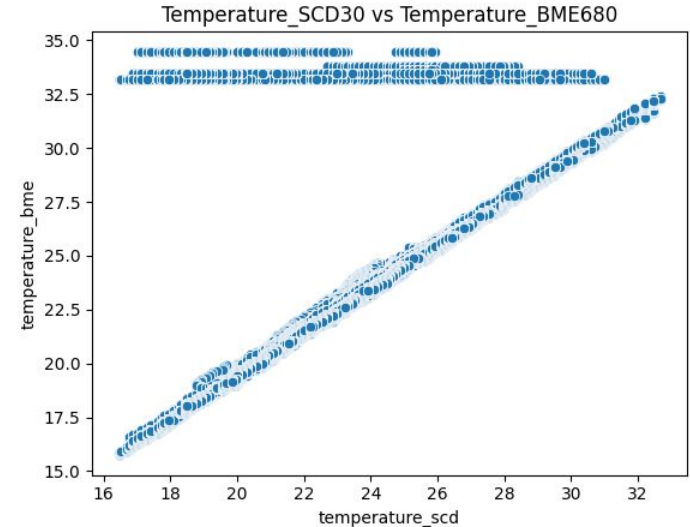
Cross-Sensor Analysis

- Analyzed inter-sensor correlation between SCD30 and BME680
- Both sensors record humidity and temperature → expected high correlation
- Humidity correlation is high: +0.97
- Temperature correlation unexpectedly low: +0.35



Cross-Sensor Analysis

- Scatter plot reveals
 - Clear linear cluster for "normal" readings
 - Flat region in temperature_bme between 32.5–35°C → likely sensor stuck/overheating
 - Anomalous readings account for only 18.77% of records

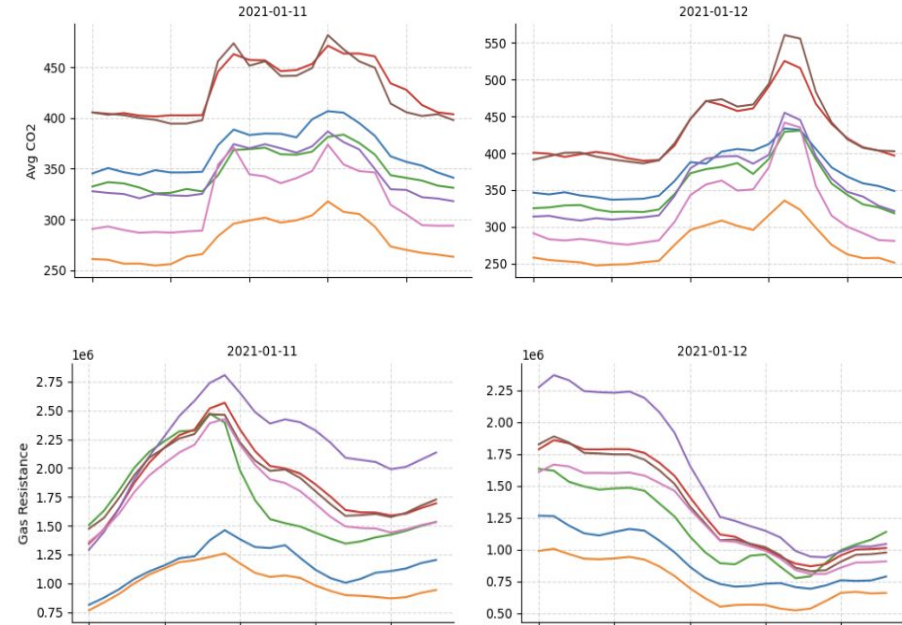


Cross-Sensor Analysis

- High humidity correlation and linear relationship of temperature readings suggest redundancy
- Main signal from SCD30: CO₂ measurements
- Explored relationship between CO₂ (SCD30) and BME680 readings
- BME680 gas resistance reflects VOC concentration (negatively correlated) [1]
- VOCs often share temporal trends with CO₂ [2]

Cross-Sensor Analysis

- Hypothesis: CO₂ can be estimated using BME680 sensor data
- Observed negative correlation between CO₂ and gas_resistance
- Developed multiple models to predict CO₂ from BME680 features
 - Reduce sensor redundancy
 - Possibly omit SCD30 sensor without significant loss of information

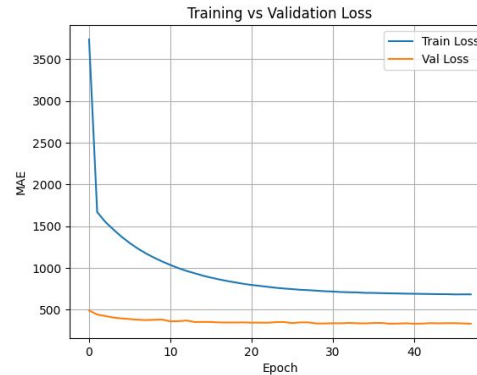


Cross-Sensor Analysis

- **Model A** (X = time fields, gas_resistance) and **Model B** (X = time fields, all readings of BME680)
- Performance on test split:

Tolerance level	Model A	Model B
$\pm 5 \%$	73.05%	76.19%
$\pm 10 \%$	94.17%	95.91%

- Limitation: Could not validate on unseen data (2023) due to restricted IMF lab access



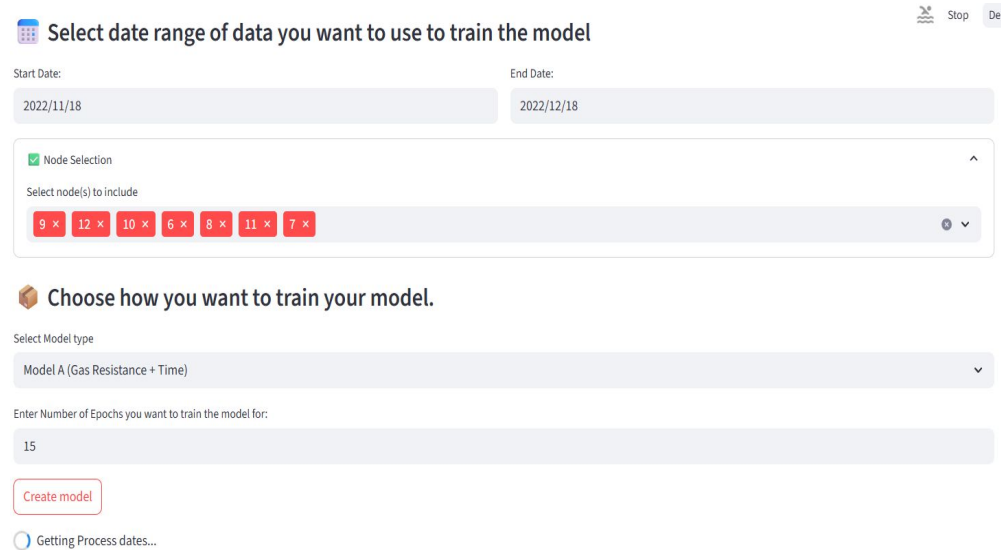
Model A



Model B

Cross-Sensor Analysis

- Users can:
 - Use Pretrained model A or model B
 - Train their own model with custom date range, nodeIDs
 - Can also mention number of epochs
- After training, user can:
 - Download model and scaler
 - Reuse the latest trained model and scaler for future predictions
- System stores most recent model for each type



The screenshot shows a web interface for training a model. At the top right, there are 'Stop' and 'Dej' buttons. The main heading is 'Select date range of data you want to use to train the model'. Below this, there are two date input fields: 'Start Date:' with the value '2022/11/18' and 'End Date:' with the value '2022/12/18'. A 'Node Selection' section follows, with a checked checkbox and the text 'Select node(s) to include'. Below this, there is a row of red buttons with white text: '9 x', '12 x', '10 x', '6 x', '8 x', '11 x', and '7 x'. A dropdown arrow is visible on the right. The next section is 'Choose how you want to train your model.' with a 'Select Model type' dropdown menu showing 'Model A (Gas Resistance + Time)'. Below this is an 'Enter Number of Epochs you want to train the model for:' input field with the value '15'. A 'Create model' button is located below the epoch input. At the bottom, there is a loading indicator with the text 'Getting Process dates...'.

Cross-Sensor Analysis

- Users can evaluate model performance in 3 ways:

1) Manual single-row input

- Predict CO₂ for user-provided feature vector

Entering Details for a Single Row ▼ Depi

 **Provide Input for Prediction**

Select Node ID

6 ▼

 **Enter Timestamp and Node Info**

Select Date

2021/11/15

Enter Hour (0-23)

10

Enter gas resistance.

450424

☒ Predict CO₂ Level

 Predicted CO₂ Level: 375.89 ppm

Cross-Sensor Analysis

- Users can evaluate model performance in 3 ways:

2) CSV batch upload

- Upload dataset and get CO₂ predictions
- Downloadable output with predicted values

Upload CSV for Batch Prediction

Upload a CSV file with a 'timestamp', 'gas_resistance', and 'node_id' column



Drag and drop file here

Limit 200MB per file • CSV

Browse files



example.csv 448.0B



✓ Predictions completed!

		month_cos	week_number_sin	week_number_cos	node_6	node_7	node_8	node_9	node_10	node_11	node_12	predicted_co2
0	002	1	-0.1205	0.9927	☐	☒	☐	☐	☐	☐	☐	306.9866
1	002	1	-0.1205	0.9927	☐	☐	☐	☒	☐	☐	☐	552.7454
2	002	1	-0.1205	0.9927	☒	☐	☐	☐	☐	☐	☐	439.1749
3	002	1	-0.1205	0.9927	☐	☐	☐	☐	☐	☒	☐	535.4441
4	002	1	-0.1205	0.9927	☐	☐	☒	☐	☐	☐	☐	412.2923



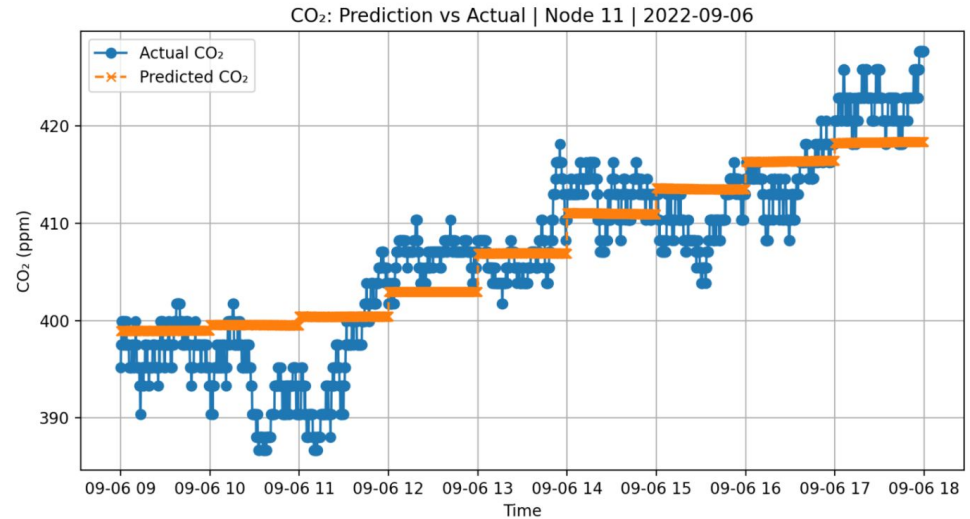
Download CSV with Predictions

Cross-Sensor Analysis

- Users can evaluate model performance in 3 ways:

3) Database-based test

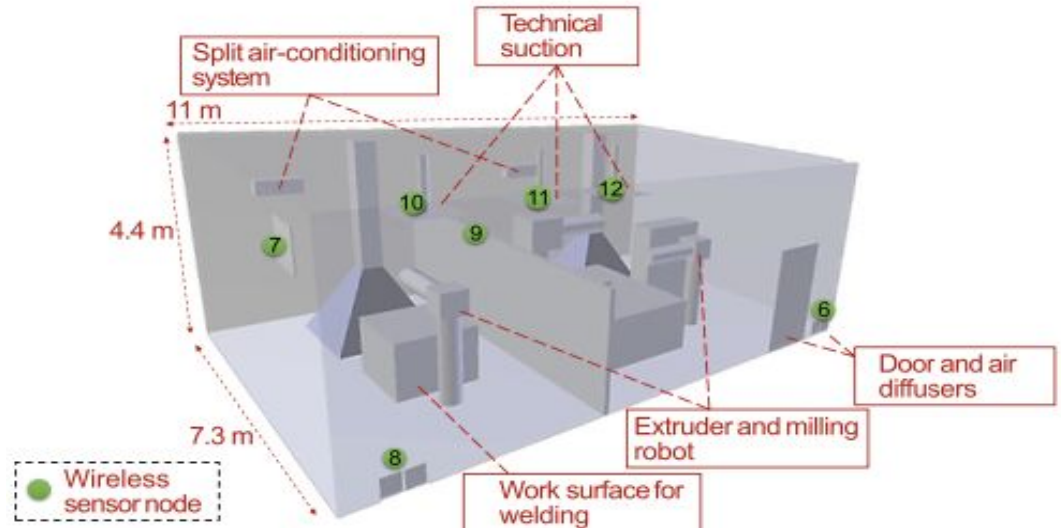
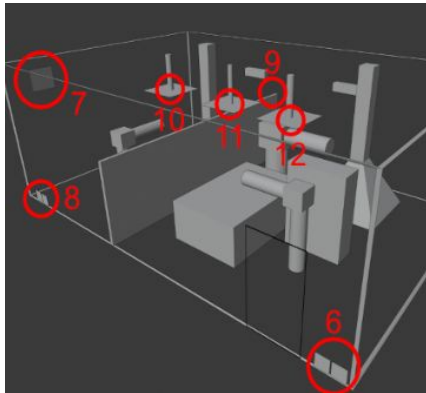
- Select date and node_id
- Visual comparison of predicted vs actual CO₂



Node Analysis

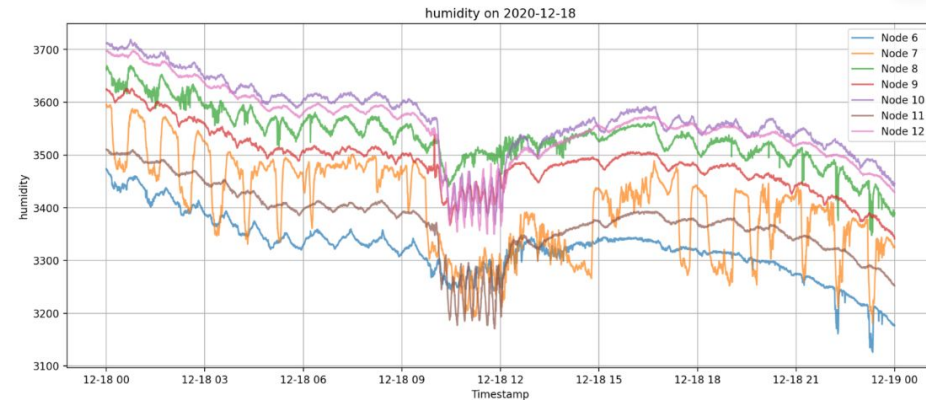
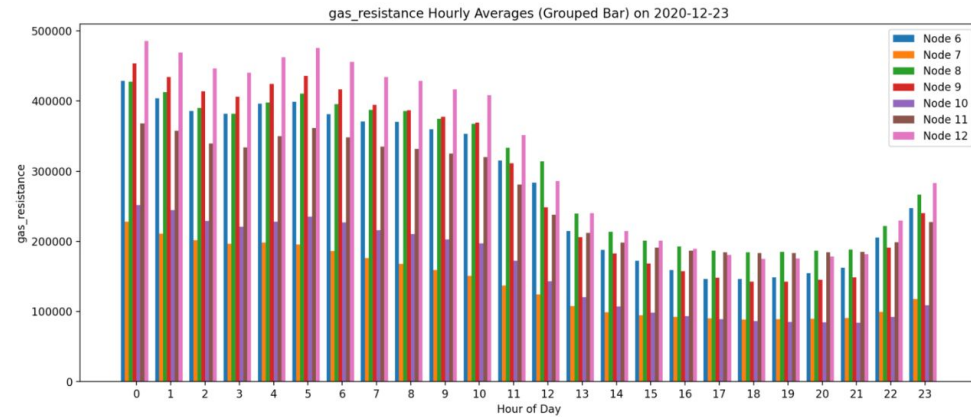
Node Analysis

- Behavioral Insights and Reliability
- Detect anomalies and deviations
- Assess redundancy
- Uncover behavioral grouping



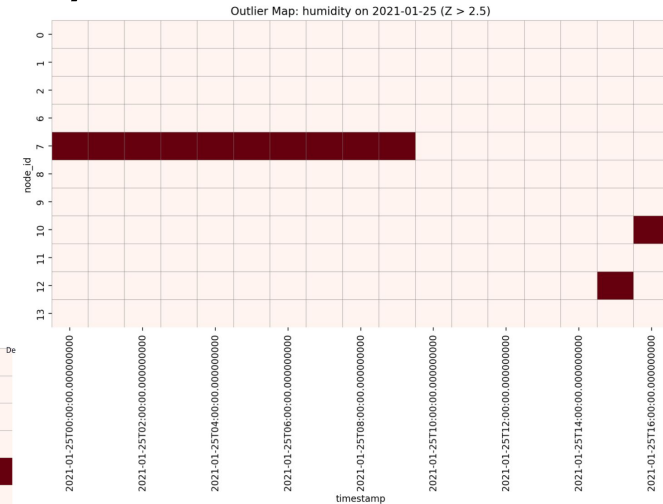
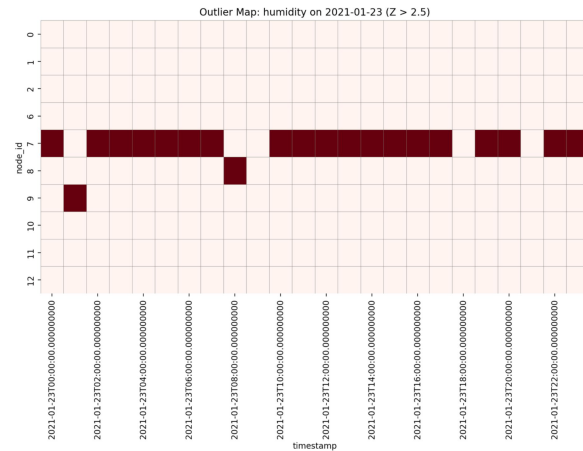
Node Analysis: Trend Viewer

- Visualize trends through multiple plot types
 - Line
 - Bar
 - Heatmap
 - Multi-day trends



Node Analysis: Anomaly Detection (Z-Score)

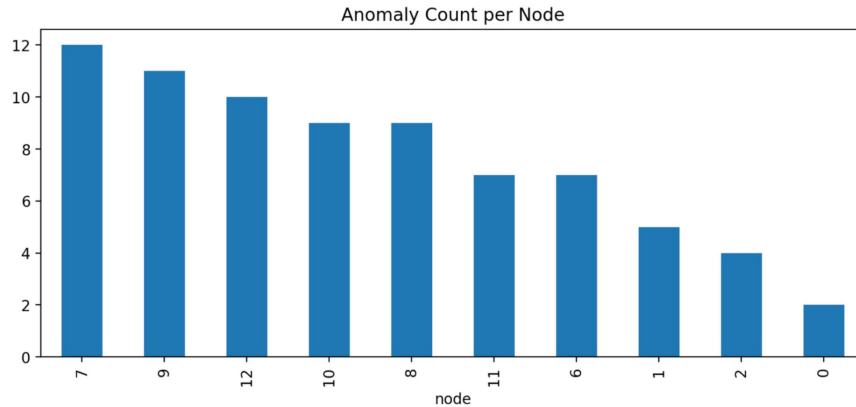
- Z-Score: detects hourly deviations from node average
- Customisable by feature, date and threshold
- Visual output:
 - Outlier heatmap
 - Node Anomaly Count
 - Deviation Frequency



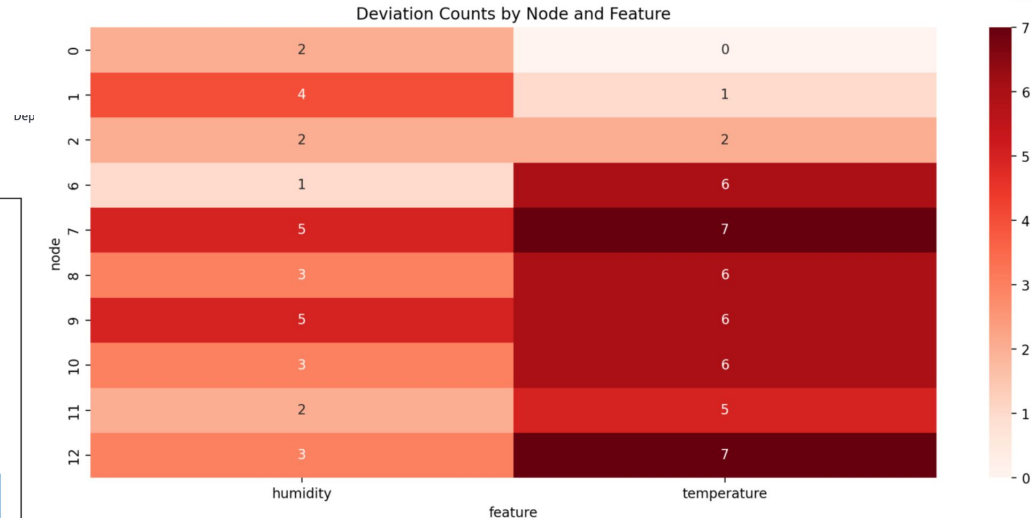
Node Analysis: Monthly Deviation Summary

- January 2021:
 - Node 7 with highest anomaly counts
 - Followed by nodes 9, 12, 10, etc.

 Anomaly Count per Node

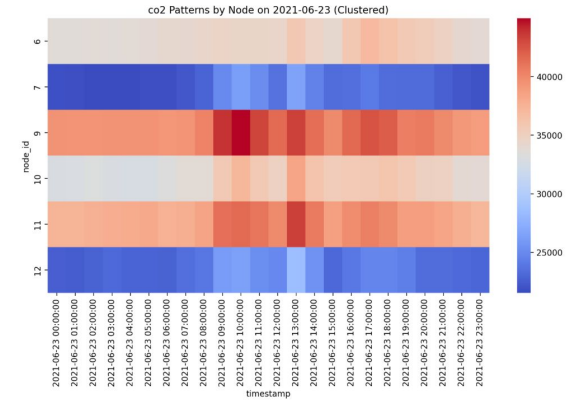
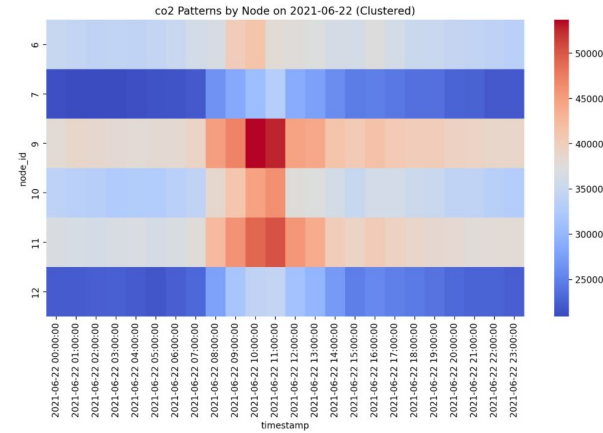


 Deviation Counts by Node and Feature



Node Analysis: Clustering Behavior

- K-Means group nodes with similar trends
- Stable clusters suggest
 - Physical proximity
 - Similar surroundings (airflow, occupancy conditions)
- Repeated clustering could indicate redundancy



Cluster Assignments

	node_id	cluster
0	9	0
1	11	0
2	7	1
3	12	1
4	6	2
5	10	2

Cluster Assignments

	node_id	cluster
0	9	0
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3	12	1
4	6	2
5	10	2

Thank you.

Reference

W. M. Rand (1971). "Objective criteria for the evaluation of clustering methods". Journal of the American Statistical Association. 66 (336). American Statistical Association: 846–850. doi:10.2307/2284239. JSTOR 2284239.

[1] Luis Llamas. Environmental quality sensor with arduino and bme680.
<https://www.luisllamas.es/en/environmental-quality-sensor-arduino-bme680/>, 2020. Accessed: 2025-07-14.

[2] Betty Molinier, Caleb Arata, Erin F Katz, David M Lunderberg, Jennifer Ofodile, Brett C Singer, William W Nazaroff, and Allen H Goldstein. Bedroom concentrations and emissions of volatile organic compounds during sleep. Environmental Science & Technology, 58(18):7958–7967, 2024.